

Presentation meeting

Mathieu Ouillon

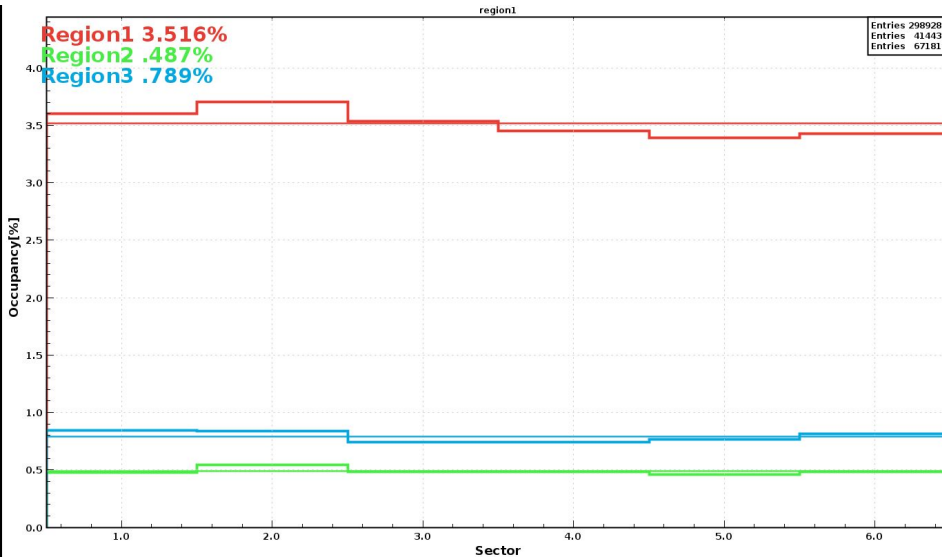
08/20/24

Mississippi State University

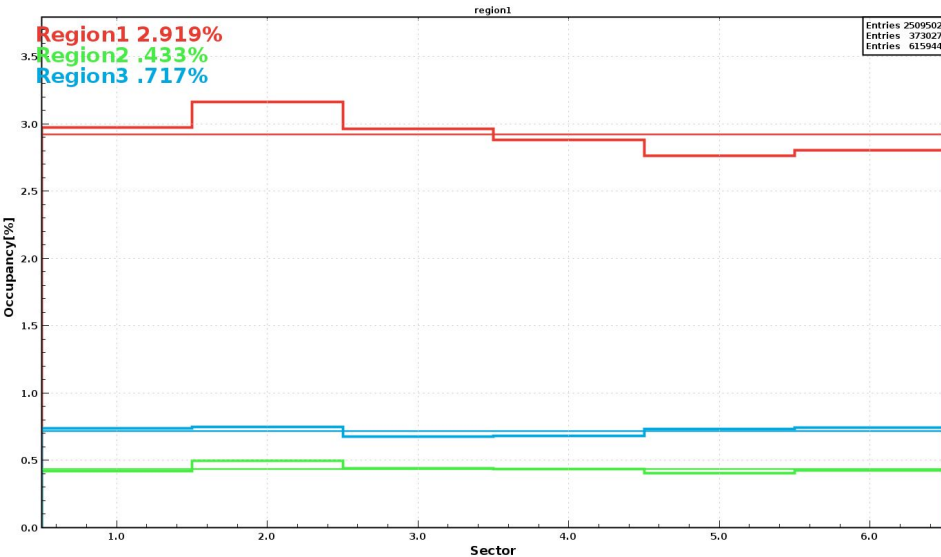


MISSISSIPPI STATE
UNIVERSITY™

DC occupancy

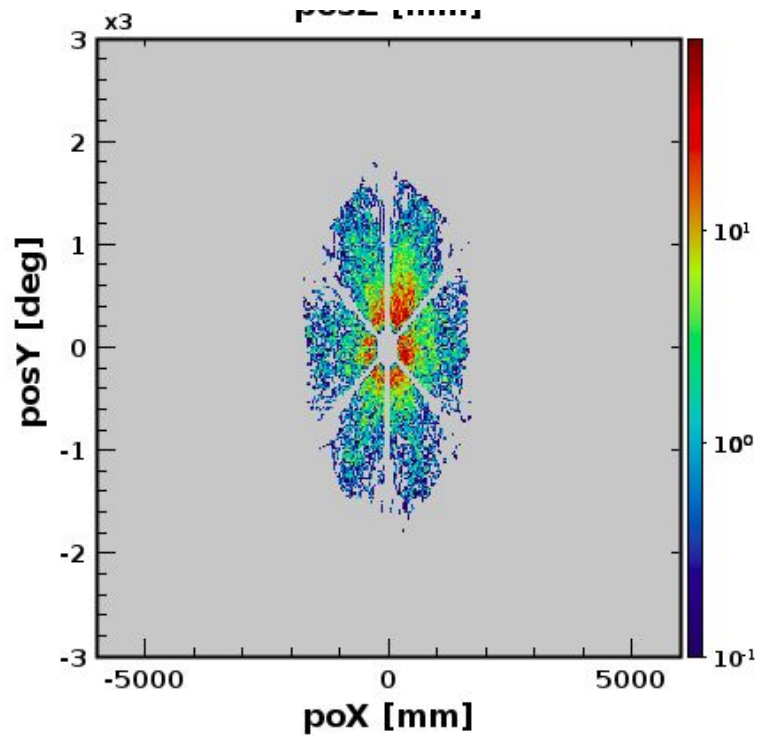


ALERT_FTO n

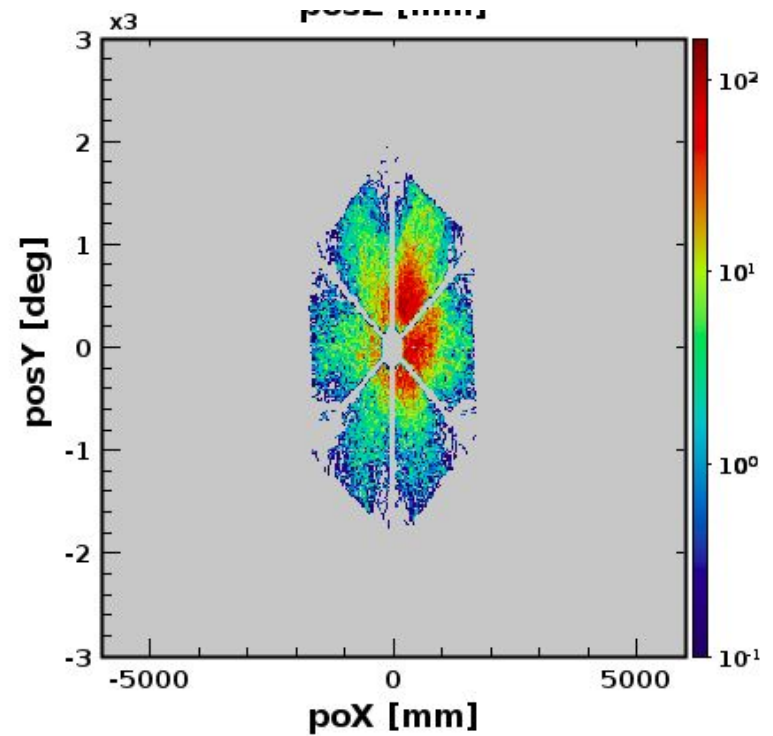


RG-A_FTO n

DC position of background



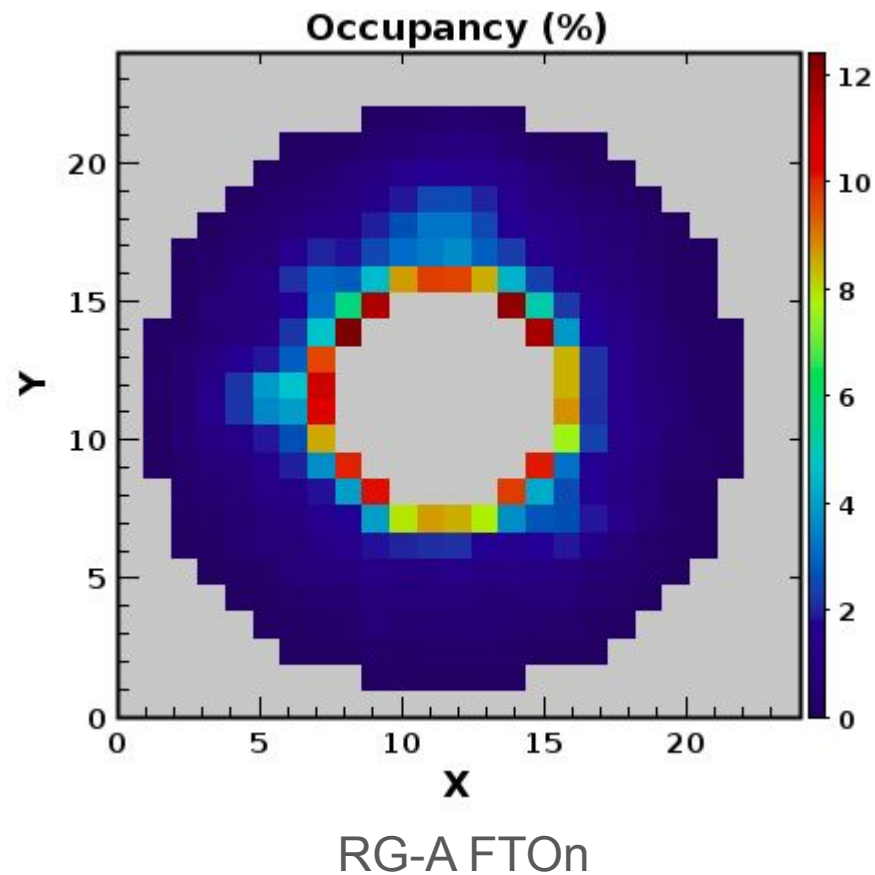
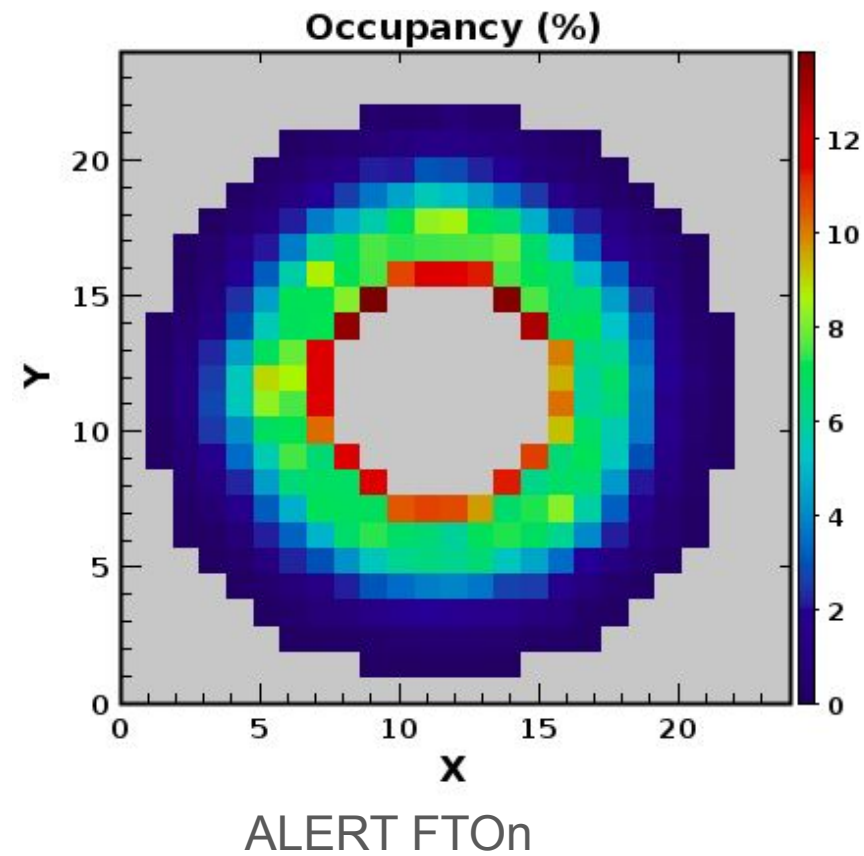
ALERT FTOn



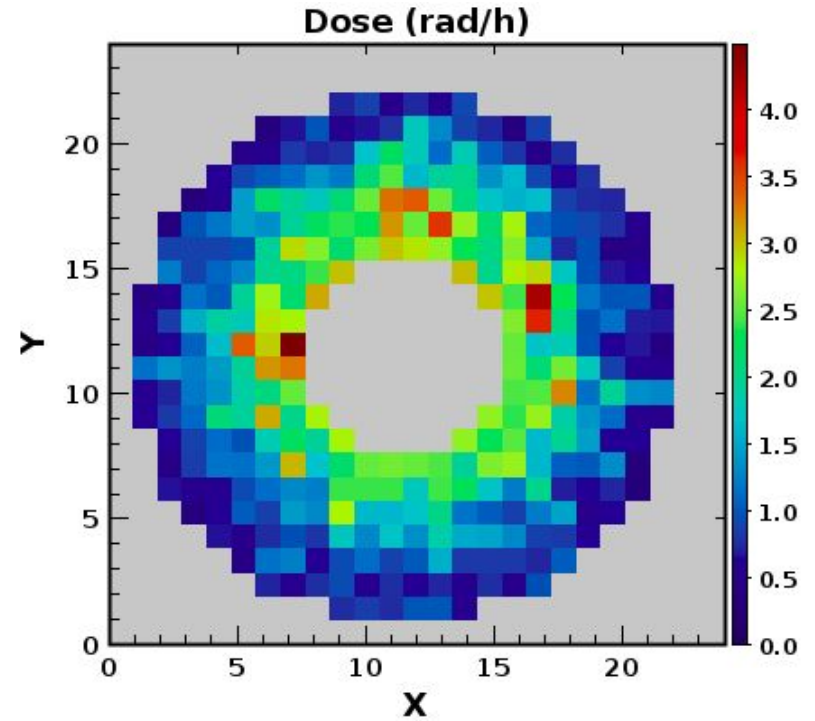
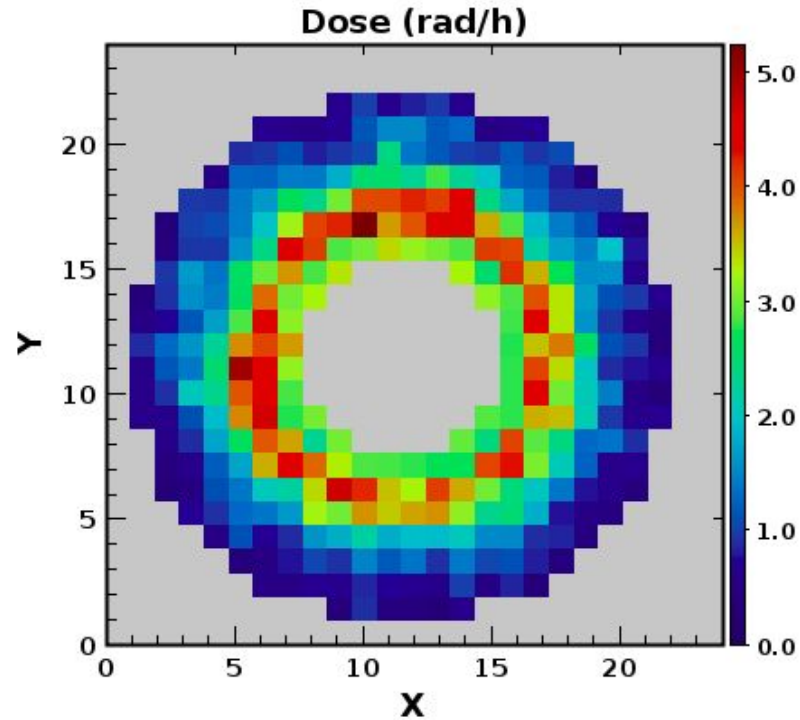
RG-A FTOn

Found asymmetry in DC. Need to find why ?

FT Occupancy



FT Dose



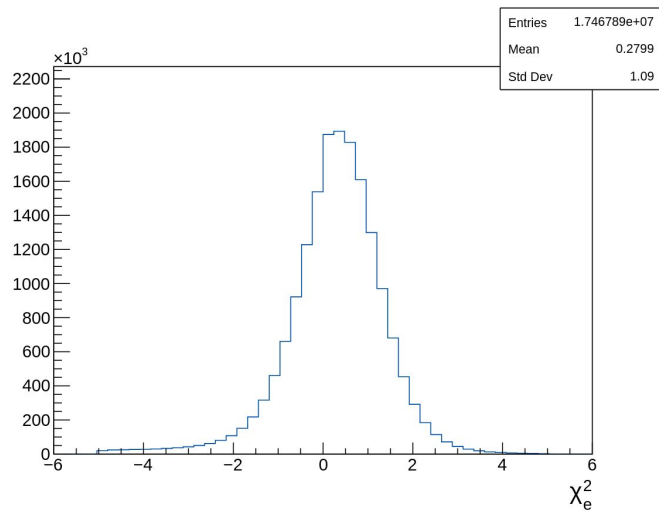
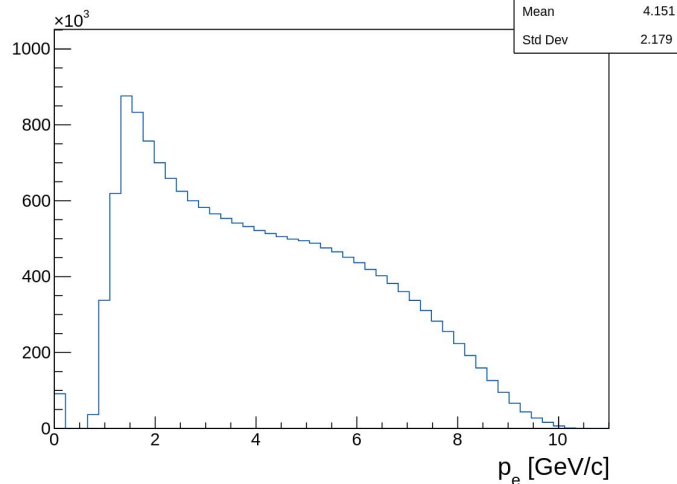
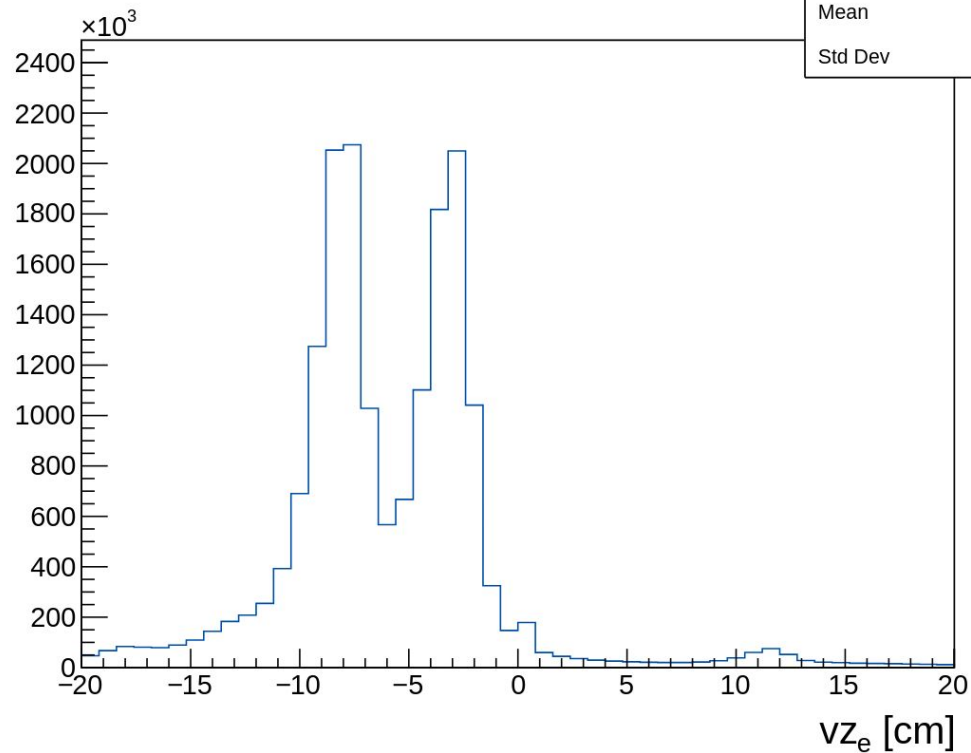
Summary of ALERT study

- So we have a higher occupancy in DC compared to the study done by R. De Vita in 2017 (probably due to the new inert material added recently).
- Found an asymmetry in DC background. Work on that.
- FTCal occupancy is around 12%.
- Dose for ALERT with FT On is around 5 rad/h.
- As the RG-C studies have shown, due to a drop of light yield, ALERT group needs to keep the dose close to 2.5/3 rad/h to keep a good reconstruction.
- A possible solution is to reduce luminosity.

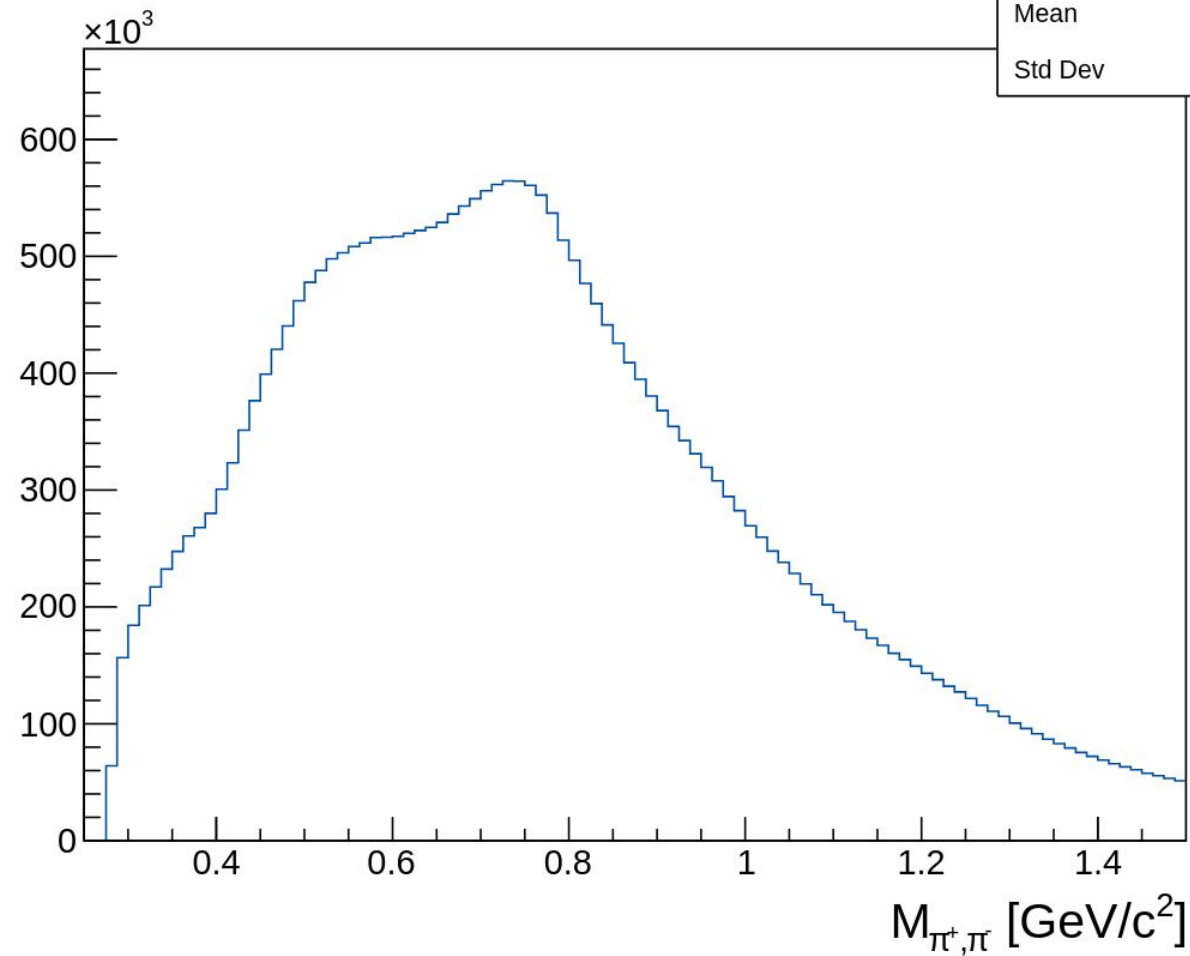
Analysis procedure

- Use CxC run : 018451 and 018452
 - from /cache/hallb/scratch/rg-d/production/prod/v4ob_aideCxC/dst/recon/018451
 - Find the trigger electron : status < 0
 - status : a summary of the particle's detector "topology", **with a negative sign to designate the trigger particle**
 - https://clasweb.jlab.org/wiki/index.php/CLAS12_DSTs
 - Find every combination of at least one π^+ and one π^- .
 - Plot the invariant mass of all combinations.
 - nb electron : 17.46789M | nb π^+ : 23.83272M | nb π^- : 22.00181M
 - Compare to Matt : e- : 17.37665M | nb π^+ : 23.67921M | nb π^- : 21.88022M
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- Only cut on vz_electron : -20 <= vz <= 20
 - Same of the pions.

Electron kinematics



Rho mass distribution



Non-work topics

- Apply for my SSN : Should have it in two weeks